Resume Project Demo

Distributed Key-Value Store

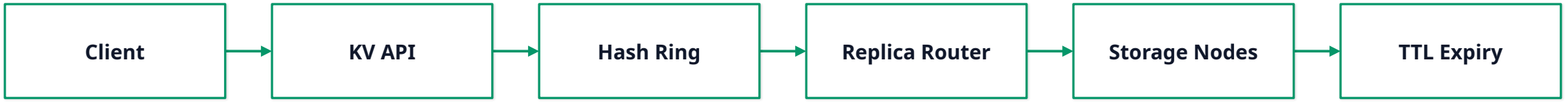
Consistent hashing, replication, TTL and simulated node failure

Built to explain backend engineering, data structures, failure handling, APIs and system design in an SDE I interview.

Demo-ready
Code + Tests + Docker + PDFs

Problem and Architecture

A single storage node is not enough when systems require routing, replication and availability.



The design separates request handling, core logic and reliability concerns. That makes it easier to explain scaling and failure handling during interviews.

Key idea

Keep the demo simple enough to run locally, but design it like a real backend service with production concerns.

Core Engineering Features

What this project proves on your resume

- GET / PUT / DELETE APIs
- Consistent hashing
- Replication factor two
- TTL support
- Node down/up simulation

This project is useful because each feature maps to a real SDE interview topic: data structures, APIs, concurrency, testing, state management, and scale.

Use the feature list to lead the explanation. Do not say only "I built an app". Say what engineering problem each feature solves.

API	**Tests**	**Demo**	**Deploy**
REST	pytest	Browser	Docker

Live Demo Flow

A simple sequence to show the project confidently

1. PUT a key
2. GET from routed node
3. Inspect replicas
4. Take node down

Presenter script

1. Start with the problem.
2. Show the architecture.
3. Run the browser demo.
4. Open the API response.
5. End with what you learned.

Command

```
pip install -r requirements.txt && uvicorn app.main:app --reload --port 8000
```

Interview Talking Points

Use these points to answer project questions clearly

1. Hashing routes data
2. Replication improves availability
3. TTL cleans temporary keys
4. Node failure changes routing behavior

Resume impact

This project is not just a CRUD app. It shows backend correctness, system design thinking, automated tests and demo ownership.

Best closing line: "I can extend this with Redis, Kafka/SQS, real database migrations, metrics, tracing and AWS deployment."